

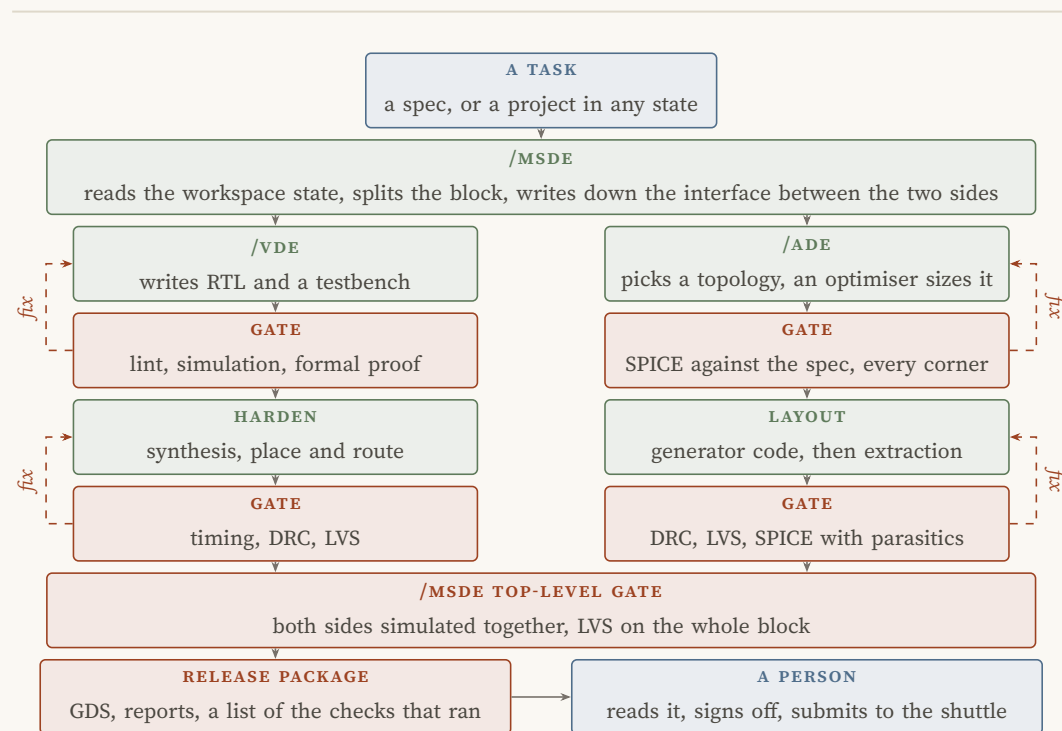
A chip design flow

Three skills that work the way /hwde does. It's a proposal, and nothing is built yet.

/vde designs digital blocks in Verilog. /ade designs analog blocks. /msde takes a mixed-signal task, splits it, and drives the other two.

Each one takes a task wherever the project stands, so you can ask it to start from a spec, review a block, fix a timing path, resize an amplifier, or carry on where it stopped. The full run from a spec to a layout ready to submit is one of those tasks.

FIGURE 1 *one task, from a spec to a signed-off block*



When a gate fails, the work goes back to a fixer agent with a set number of tries, then to a person.

Slate is a task or a person, sage is an agent doing design work, and rust is a script that passes or fails.

Figure 1 is the proposed flow. Nothing in it is built yet. September 2026.

What carries over from /hwde

The method carries over whole, but the tools, the checks and the design knowledge are all new.

TABLE 1 *what /hwde does today, and what the chip flow keeps*

	/HWDE TODAY	THE CHIP FLOW
Workspace	One folder per board, with a state.json. A killed session resumes from that file alone.	One folder per chip block, the same file, the same resume.
Front door	A task router with 17 verbs. The full run is one of them.	The same router. Verbs such as review, fix-timing, resize, add-corner, harden, resume.
Design work	23 subagent roles. The orchestrator never opens a design file.	The same rule. New roles: RTL writer, testbench writer, analog designer, layout writer, reviewer, fixer.
Checks	62 scripts with one contract: JSON out, exit 0, 1 or 2, no prompts.	The same contract, wrapped around the open tools on the next page.
Gates	A release gate refuses unless every check that applies has run.	The same rule. The report lists what ran, not what was planned.
Stale work	An edit marks the gates it invalidated. Input hashes catch the rest.	The same map. An RTL edit stales simulation, synthesis, layout and signoff.
Fix loop	Snapshot, one fixer per cluster of findings, a budget, then a person.	The same.
Learning	LEARNINGS.md has 339 dated entries. Each stage has a bench on frozen inputs.	The same files, starting empty.
Money	Only a person can pay for boards.	Only a person can submit to a shuttle.

Four things have no equivalent in /hwde

FEEDBACK IS SLOW

A board comes back in weeks, but a chip takes 6 to 9 months to make.¹ So simulation carries more weight, and the flow has to test the testbench itself.

RUNS ARE LONG

Hardening a block takes far longer than a board DRC, so a gate becomes a job that records its result in state.json instead of a call the agent waits on.

ANALOG LAYOUT HAS NO AUTOROUTER

So the agent doesn't draw. It writes code that generates the layout, and the checkers judge the result. I think this is the least proven part of the flow.

THE PDK IS A PINNED DEPENDENCY

The way the KiCad version is in /hwde, and the flow would start with one.

The tools, next to ECE 298A

Your course flow is the digital half of this already. The flow would keep all of it, run it on the box, and add the checks and the analog side.

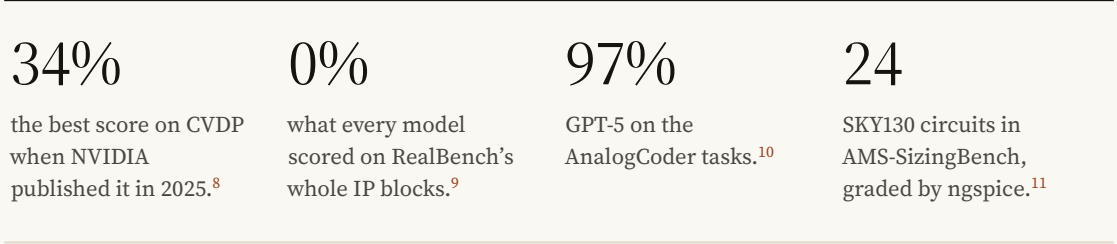
TABLE 2 *what ECE 298A uses, and what the chip flow would use*

	ECE 298A USES	THE CHIP FLOW
Simulate	Verilog, cocotb tests, Icarus.	Kept. Verilator is added for lint and for long runs.
Prove	Nothing.	SymbiYosys. Formal proof finds what a testbench misses.
Harden	LibreLane, inside Tiny Tapeout's GitHub Action. ²	Kept, with the same config file. It runs on the box, so the agent reads the logs in minutes and without a push.
Sign off	The Tiny Tapeout precheck. It runs DRC and no LVS. ³	Kept. Magic and KLayout DRC, netgen LVS and OpenSTA timing are added, on every design.
Analog	Not in the course flow. Tiny Tapeout's analog path is done by hand.	ngspice driven from Python. Layout from generator code with gdsfactory and gLayout. Magic, KLayout and netgen as checkers only.
Both at once	Nothing.	ngspice simulating together with Icarus or Verilator. ngspice ships the glue. ⁴ It is thinly documented.
PDK	GF180MCU.	The same first, so course work drops straight in. SKY130 and IHP SG13G2 are the other two open ones.
Shuttle	Tiny Tapeout.	The same. A tile is listed at 70 EUR. ⁵ wafer.space sells a larger GF180MCU slot from 2,000 USD. ⁶
Install	GitHub's servers.	One container, IIC-OSIC-TOOLS, holds nearly every tool above and all three PDKs. ⁷

Every tool named here had a release or a commit in 2026, except gLayout. Its home repository has been quiet for about a year, so an active fork carries it. OpenLane itself is retired, and LibreLane took over. The FOSSi Foundation has kept it since Efabless closed in 2025.²

How you would know it works

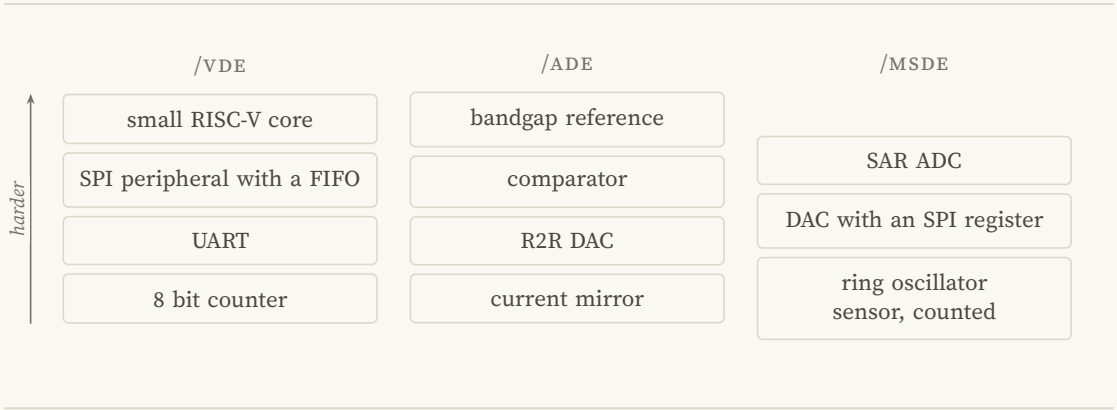
Public benchmarks show how the agents compare with everyone else’s, but only the flow’s own corpus says whether you can trust a finished design.



CVDP has 783 RTL design and verification problems, graded with open tools. The older sets, VerilogEval and RTLLM, are close to solved.¹² The best score on RealBench’s single modules was 13%, and on ChipBench, from 2026, the best model has 31%.¹³ The AnalogCoder tasks use ideal devices and no PDK, so that 97% says little about a real amplifier. AMS-SizingBench is the closest public test to /ade, and it covers sizing only, not layout.

The flow’s own corpus is a ladder of reference designs, and a rung counts when every gate on page 1 is green for it. Each design also gets faults planted in it, such as a removed reset, a shorted net or a swapped device size, and a gate that misses its planted fault is broken. /hwde does this today with three golden boards and seven mutants.

FIGURE 2 the flow’s own corpus, one column per skill



Each column is one skill’s reference designs, and the arrow points to the harder ones.

Scores as sources 8 to 13 published them. Figure 2 is a plan: nothing in it is built yet.

What a pass means. The design meets its written spec in simulation at the corners it declares, it breaks no rule of the PDK, and its layout matches its schematic.

WHAT IT DOESN'T MEAN It doesn't say the silicon works. The spec can be wrong, the testbench can miss a case, and the device models can be off. Only a measured chip settles it.

Where the field stands

Agents now write modules that work, but nobody has shown one taking a chip to working silicon on its own.

What the field has. Synopsys and Cadence sell design agents that drive their own licensed tools, and Cadence launched a “Super Agent” for verification in February 2026.¹⁴ Every performance figure I found for these products came from the vendor. In research, an agent took a 219 word spec to a finished RISC-V layout in 12 hours, though the PDK it used can't be fabricated and the paper isn't peer reviewed.¹⁵ Claude Code, which this flow would run on, scored 63% on CVDP's agent tasks in July 2026, in a single run.¹⁶ A university group already uses Claude Code skills to port designs through OpenROAD.¹⁷

What it doesn't have. The first chip with LLM written Verilog went out on Tiny Tapeout in 2023, but a person wrote its tests and ran the flow.¹⁸ The only agent output I found measured in silicon is a ring of I/O pads around a core made by hand.¹⁹ Tests pass too easily. Of the generated code that passed one benchmark's testbenches, 44% then failed formal verification.⁹ An agent loop can make results worse. One frontier model fell from 59% to 42% on CVDP when wrapped in one, and most of those failures were silent, because the loop finished and the answer was wrong.²⁰ Code that works is not yet good code, and on one benchmark it reached 64% of an expert's area efficiency.²¹

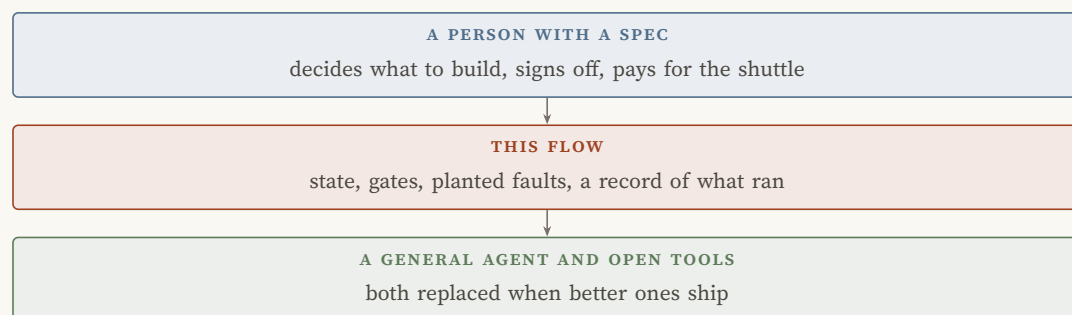
It bets, as /hwde does, on a general agent with open tools and checks that are hard to fool.

THE ONE CLAIM THIS BRIEF MAKES

WHAT THIS DOES NOT MEAN It doesn't try to build a better model or compete with the vendors. A person still signs off, and it stays small, at one Tiny Tapeout tile.

Where this flow sits. Formal proof and planted faults answer the tests that pass too easily, and a record of which checks ran answers the silent failures. I found no published open flow that covers digital and analog with LVS in the loop, so that part would be new.

FIGURE 3 *the three layers, and which one this flow is*



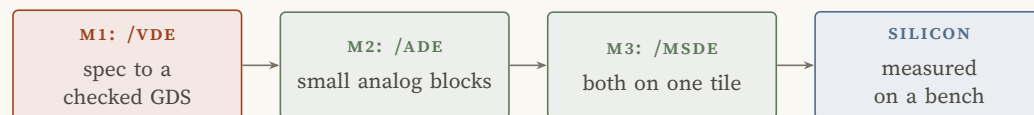
The slate box is the person, the rust box is this flow's checks, and the sage box is the agent and tools it drives.

Figure 3 carries no measured number. It shows the proposal as of September 2026.

A first milestone

Build /vde first, on GF180MCU, on the box. It's the part your course needs this term, and it builds the skeleton that /ade and /msde would reuse.

FIGURE 4 *the milestones, and where silicon comes in*



The rust box is the first milestone, the one this page asks for. Sage boxes are the milestones after it, and the slate box is the chip on a person's bench.

What M1 builds. It pins the tools into a container and ports the /hwde skeleton over: state, router, gate runner, stale map and fix loop. It adds seven gates for lint, simulation, formal, harden, timing, DRC with LVS and gate-level simulation, and it plants faults in three reference designs.

M1 is done when the counter, the UART and the SPI block each go from a written spec to a GDS that passes every gate with no hand edits, every planted fault is caught, your course counter passes a review task, and a CVDP run has given a number to compare with page 4.

TABLE 3 *what M1 would take*

AMOUNT	WHAT IT IS
About 3 weeks	of box time, mostly unattended. /hwde v1 took 15 steps over 23 days. I estimate about 10 steps here, because the skeleton exists.
\$500 to \$1,000	of model usage. This is a guess, not a measurement. Tracks on this box cost 3 USD at the median and 100 USD at most, and none of them was an EDA build.
70 EUR	is the listed price of a Tiny Tapeout tile, if you want silicon from it. ⁵ The tools and the PDK cost nothing.
6 to 9 months	to manufacture a shuttle, ¹ and up to a year before chips arrive. Nothing in M1 waits on that.

Three decisions are yours

1. *Which PDK first.* GF180MCU matches the course. wafer.space sells slots on it now, and Tiny Tapeout has run GF shuttles, though I couldn't confirm the next one. SKY130 has the most analog examples and the one public analog benchmark, and IHP has the only Tiny Tapeout shuttle open today. I'd start with GF180MCU.

2. *Digital first or analog first.* Digital is where agents are strongest and the loop is fast, but analog is closer to research. I'd go digital first, and I wouldn't put anything with a deadline behind /ade.

3. *Whether M1 aims at real silicon.* A tile is cheap, but the wait is most of a year, so an early submission starts the clock. I'd aim M1 at a tile that's ready to submit, and whether you pay for it stays your call.

Table 3's first two rows are estimates from /hwde v1's record and this box's track costs; the last two are from sources 5 and 1.

Sources

Sources 1 to 7, 9 and 14 to 21 were read on 21 September 2026.

- 1 Tiny Tapeout FAQ. tinytapeout.com/faq
- 2 LibreLane and Tiny Tapeout's GitHub Action.
github.com/librelane/librelane
github.com/TinyTapeout/tt-gds-action
- 3 Precheck source. github.com/TinyTapeout/tt-support-tools/blob/main/precheck/precheck.py
- 4 In the ngspice source tree under `src/xspice/verilog`. sourceforge.net/p/ngspice/ngspice
- 5 tinytapeout.com/specs/analog
- 6 wafer.space
- 7 github.com/iic-jku/IIC-0SIC-T00LS
- 8 Pinckney et al., NVIDIA, June 2025. arxiv.org/abs/2506.14074
- 9 RealBench, July 2025. Model: o1-preview. arxiv.org/abs/2507.16200
- 10 AnalogAgent, March 2026. arxiv.org/abs/2603.23910
- 11 AutoSizer, February 2026. arxiv.org/abs/2602.02849
- 12 RTL-BenchLS, June 2026, on why a harder set was needed. arxiv.org/abs/2606.08976
- 13 ChipBench, January 2026. Model: Claude Opus 4.5. arxiv.org/abs/2601.21448
- 14 Synopsys, and Cadence results, 17 February 2026.
synopsys.com/blogs/chip-design/agentengineer-technology-transforming-engineering-workflows.html
sec.gov/Archives/edgar/data/813672/000081367226000013/cdns02172026ex9901.htm
- 15 Design Conductor, Verkor, February 2026. The PDK is ASAP7. arxiv.org/abs/2603.08716
- 16 Si2 leaderboard, CVDP 1.1, heavy context, Claude Opus 4.8. lbc-bench.si2.org
- 17 HighTide, June 2026. arxiv.org/abs/2606.04126
- 18 Chip-Chat, May 2023. arxiv.org/abs/2305.13243
- 19 AMS-IO-Agent, December 2025, 28 nm. arxiv.org/abs/2512.21613
- 20 Yubeaton, Garg, Hegde, March 2026. Gemini 3.1 Pro. arxiv.org/abs/2603.19347
- 21 Pluto, October 2025. arxiv.org/abs/2510.14756